

Same answer, different tutor behavior

Frozen Dijkstra task - graph and answer fixed - screenshots show the audited behavior boundary.

a AlgoTutorGen - learner loop

9/9 MACHINE OK

Relax edge B→C: $nd = \text{dist}['B'] + 1 = 3$. Since $3 < \text{current dist}['C']$ (5), update $\text{dist}['C']$ to 3.
The path A→B→C (cost 3) is shorter than the direct A→C (5); Dijkstra correctly improves the distance.

graph

graph frontier / relax / path
frontier graph frontier dist

DP current call / dependency window

$\text{dist}['C']$

$nd = d + w = 2 + 1 = 3$

graph node indicator / frontier

A: $\text{dist}=0$ B: $\text{dist}=2$
C: $\text{dist}=3$

$nd = d + w = 2 + 1 = 3$

auxiliary state

original state evidence (default: do not enter main stage)

Previous Play Next

24 / 38

Main loop Main loop Main loop Main loop Main loop Main loop Main loop Main loop

graph

3 field - ("nodes":
["A", "B", "C"], "edges":
[("A", "B", 2), ("A", "C", 5), ("B", "C", 1)],
("A", "A", 0), ("B", "B", 0), ("C", "C", 0))

Frame 24/38 - predict $\text{dist}['C'] = 3$ - submit - receive correct feedback

Verified state, checkpoint, feedback, and learning record remain on one page.

b Direct HTML - 2/9

Visible before learner action

Dijkstra Shortest Path

Emergency dispatch: Find the shortest travel time from the starting intersection to all reachable intersections.

Computing shortest paths...

Expected: ("A": 0, "B": 2, "C": 3)

Stop

Sort all the nodes

Aut

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

18

19

20

21

22

23

24

25

26

27

28

29

30

31

32

33

34

35

36

37

38

39

40

41

42

43

44

45

46

47

48

49

50

51

52

53

54

55

56

57

58

59

60

61

62

63

64

65

66

67

68

69

70

71

72

73

74

75

76

77

78

79

80

81

82

83

84

85

86

87

88

89

90

91

92

93

94

95

96

97

98

99

100

Expected answer is already exposed

Static visibility is not evidence of a reachable teaching loop.

Audit: interaction + feedback chain FAIL

c Direct-BR (max 5) - 7/9

Checkpoint reached; partial behavior

GRAPH & STATE VISUALIZATION

KNN-HEAP
[("A", 0), ("B", 2), ("C", 5)]

CURRENT STATE (JSON)

```
{
  "dist": {
    "A": 0,
    "B": 2,
    "C": 5
  },
  "heap": [
    ("A", 0),
    ("B", 2),
    ("C", 5)
  ]
}
```

TEACHING EXPLANATION

Node A is popped from the heap with the smallest distance (0). We relax its outgoing edges: the path A→B gives distance 0+2=2 (which is less than ∞), and A→C gives 0+5=5 (also less than ∞). Both improved distances are pushed to the heap. Notice that C currently shows distance 5—but there might be a shorter path through B.

STEP EVIDENCE

Operation: Relax. Targets: B, C (neighbors of A).
Dependencies: dist[A]=0 must be finalized. Before: dist["A"] = 0. After: dist["A"] = 0, dist["B"] = 2, dist["C"] = 5.
Feedback state: dist["B"] = 2, dist["C"] = 5.

Checkpoint: After processing node A, what is the current distance to node C?

wrong Submit Hint

Show Answer

Incorrect. Your answer was "wrong".
Review the current state and try again.
Hint: Look at the edge weight from A to C (which is 5). Since distance to A is 0, the path through A to C has cost 5.

Wrong feedback + learning log: PASS

Correct feedback + Show answer: FAIL